Subgroups and Cyclic Groups

From the class book: Chapter 5, Sections 5.1 to 5.4 and Chapter 6, Sections 6.1 to 6.3

1. Key definitions

Subgroup. A non-empty subset H of a group G is a subgroup, $H \leq G$, if H is a group under the operation of G. $\{e\}$ and G are the trivial subgroups; any other is proper and non-trivial.

Subgroup generated by a set. $\langle S \rangle$ is the smallest subgroup containing S: all finite products of elements of S and their inverses. For one element, $\langle a \rangle = \{a^n : n \in \mathbb{Z}\}$.

Centre. $Z(G) = \{z \in G : zx = xz \text{ for all } x \in G\}$.

Centralizer. $C(a) = \{x \in G : xa = ax\}$ (some books call it the normalizer $N(a)$ of the element a).

Normalizer of a subgroup. $N(H) = \{x \in G : xHx^{-1} = H\}$.

Cyclic group. G is cyclic if $G = \langle a \rangle$ for some a, called a generator. A finite group of order n is cyclic iff it has an element of order n.

Centre, centralizers and normalizers are all subgroups; $Z(G) = \bigcap_{a \in G} C(a)$, and G is abelian iff $Z(G) = G$.

2. Main results

Theorem (Subgroup tests)
Let $\emptyset \neq H \subseteq G$.

- One-step test: $H \leq G \iff ab^{-1} \in H$ for all $a, b \in H$ (additively, $a - b \in H$).
- Two-step test: $H \leq G \iff ab \in H$ and $a^{-1} \in H$ for all $a, b \in H$.
- Finite test: if H is finite, $H \leq G \iff H$ is closed under the operation.

Proof idea for the finite test: for $a \in H$ the powers $a, a^2, \ldots$ lie in H and must repeat, so $a^n = e$ for some n, and $a^{-1} = a^{n-1} \in H$.

Theorem (Combining subgroups)
Let $H, K \leq G$.

- $H \cap K \leq G$; any intersection of subgroups is a subgroup.
- $H \cup K \leq G \iff H \subseteq K$ or $K \subseteq H$. A group is never the union of two proper subgroups.
- $HK = \{hk\}$ is a subgroup $\iff HK = KH$; this holds if G is abelian or H or K is normal.
- For finite H, K: $|HK| = \dfrac{|H|\,|K|}{|H \cap K|}$, even when HK is not a subgroup.

Theorem (Subgroups of $\mathbb{Z}$ and $\mathbb{Z}_n$)
The subgroups of $\mathbb{Z}$ are exactly $n\mathbb{Z}$, $n \geq 0$; $\langle a_1, \ldots, a_k \rangle = \gcd(a_1, \ldots, a_k)\mathbb{Z}$ and $m\mathbb{Z} \cap n\mathbb{Z} = \operatorname{lcm}(m, n)\mathbb{Z}$. The subgroups of $\mathbb{Z}_n$ are $\langle \bar{d} \rangle$, one for each divisor $d \mid n$: there are $\tau(n)$ of them.

Theorem (Basic facts on cyclic groups)

- Every cyclic group is abelian (converse fails: K_4).
- Every group of prime order is cyclic, generated by any non-identity element; a group $G \neq \{e\}$ has no proper non-trivial subgroup iff it is cyclic of prime order.
- An infinite cyclic group is isomorphic to $\mathbb{Z}$; a cyclic group of order n is isomorphic to $\mathbb{Z}_n$.
- An infinite cyclic group has exactly two generators (a and a^{-1}), and e is its only element of finite order.

Theorem (Fundamental Theorem of Cyclic Groups)

Every subgroup of a cyclic group is cyclic. If $G = \langle a \rangle$ has order n, then for each divisor d of n there is exactly one subgroup of order d, namely $\langle a^{n/d} \rangle$, and these are all the subgroups. Moreover

$$\langle a^k \rangle = \langle a^{\gcd(n,k)} \rangle, \qquad |\langle a^k \rangle| = \frac{n}{\gcd(n,k)}, \qquad [G : \langle a^k \rangle] = \gcd(n,k).$$

Corollary (Counting in a cyclic group of order n)

- a^k generates $G \iff \gcd(k,n) = 1$: there are $\varphi(n)$ generators.
- For each $d \mid n$ there are exactly $\varphi(d)$ elements of order d (and none if $d \nmid n$); hence $\sum_{d \mid n} \varphi(d) = n$.
- G has $\tau(n)$ subgroups, all cyclic.
- The number of $x \in G$ with $x^m = e$ is $\gcd(m,n)$.

3. Standard examples and tables

Subgroups and non-subgroups

- $n\mathbb{Z} \leq \mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$ (addition); $\{\pm 1\} \leq \mathbb{Q}^* \leq \mathbb{R}^* \leq \mathbb{C}^*$.
- $SL(n,\mathbb{R})$, the invertible upper triangular matrices, and $\left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} : a \neq 0 \right\}$ are subgroups of $GL(n,\mathbb{R})$ (resp. $GL(2,\mathbb{R})$).
- $\mathbb{N}$ is not a subgroup of $(\mathbb{Z},+)$ (no inverses); the odd integers are not closed; $2\mathbb{Z} \cup 3\mathbb{Z}$ is not a subgroup since $3 - 2 = 1 \notin 2\mathbb{Z} \cup 3\mathbb{Z}$.
- In S_3 the set $\{x : x^2 = e\}$ has 4 elements and is not a subgroup.

Centres of standard groups

Group	$Z(G)$	Group	$Z(G)$
abelian G	G	Q_8	$\{\pm 1\}$
S_n, $n \geq 3$	$\{e\}$	A_n, $n \geq 4$	$\{e\}$
D_n, n odd	$\{e\}$	D_n, n even	$\{e, r^{n/2}\}$
$GL(n,F)$	$\{\lambda I : \lambda \neq 0\}$	$SL(2,\mathbb{R})$	$\{\pm I\}$

Also $Z(G_1 \times G_2) = Z(G_1) \times Z(G_2)$, and $Z(G)$ is always normal. Centralizers: in S_3, $C((12)) = \{e, (12)\}$ and $C((123)) = A_3$; in Q_8, $C(i) = \{\pm 1, \pm i\}$. In general $|C(a)| = |G|/|\mathrm{cl}(a)|$.

Number of subgroups

Group	Order	Subgroups	Remarks
$\mathbb{Z}_n$	n	$\tau(n)$	one for each divisor
$\mathbb{Z}_p \times \mathbb{Z}_p$	p^2	$p+3$	$p+1$ of order p
$K_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$	4	5	three of order 2
$S_3 \cong D_3$	6	6	three of order 2, one of order 3
$\mathbb{Z}_2 \times \mathbb{Z}_4$	8	8	6 cyclic
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$	8	16	seven of order 2, seven of order 4
Q_8	8	6	all normal; $\{1\}, \langle -1\rangle, \langle i\rangle, \langle j\rangle, \langle k\rangle, Q_8$
D_n	$2n$	$\tau(n)+\sigma(n)$	D_4: 10, D_5: 8, D_6: 16, D_8: 19
A_4	12	10	$1+3+4+1+1$; no subgroup of order 6
S_4	24	30	$1+9+4+7+4+3+1+1$ (orders 1, 2, 3, 4, 6, 8, 12, 24)

In D_n: $\tau(n)$ subgroups of rotations and $\sigma(n)$ subgroups containing reflections; of all these, $\tau(n)+n$ are cyclic. The 7 subgroups of order 4 in S_4: three cyclic $\langle (a\,b\,c\,d)\rangle$, the normal V_4, and three Klein groups $\{e, (a\,b), (c\,d), (a\,b)(c\,d)\}$.

Counting in $\mathbb{Z}_n$

n	$\varphi(n)$ generators	$\tau(n)$ subgroups	elements of order d
12	4	6	$d=2$: 1, $d=4$: 2, $d=6$: 2
20	8	6	$d=4$: 2, $d=5$: 4, $d=10$: 4
24	8	8	$d=6$: 2, $d=8$: 4, $d=12$: 4
30	8	8	$d=6$: 2, $d=10$: 4, $d=15$: 8
200	80	12	$d=8$: 4, $d=25$: 20, $d=100$: 40

Generators of $\mathbb{Z}_{20}$: 1, 3, 7, 9, 11, 13, 17, 19. The subgroup lattice of $\mathbb{Z}_{12}$ is the divisor lattice of 12 turned upside down: $\langle \bar{1}\rangle$ (order 12) contains $\langle \bar{2}\rangle$ (6) and $\langle \bar{3}\rangle$ (4); then $\langle \bar{4}\rangle$ (3) and $\langle \bar{6}\rangle$ (2); then $\{\bar{0}\}$.

Cyclic and non-cyclic examples

- Cyclic: $(\mathbb{Z},+) = \langle 1\rangle = \langle -1\rangle$; $\mathbb{Z}_n$; μ_n; $\{\pm 1, \pm i\} = \langle i\rangle$; $\left\{\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}\right\}$ with $a \in \mathbb{Z}_n$ (order n) or $a \in \mathbb{Z}$ (infinite); $U(7) = \langle 3\rangle$.
- Not cyclic: $(\mathbb{Q},+)$, $(\mathbb{R},+)$, $\mathbb{Q}^*$ (abelian); $U(8)$, $U(12)$, $U(15)$; K_4.
- $K_4 = \{e, a, b, c\}$, $a^2 = b^2 = c^2 = e$, $ab = c$, $bc = a$, $ca = b$: the smallest non-cyclic group; $K_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \cong U(8) \cong U(12)$. Every group of order 4 is $\mathbb{Z}_4$ or K_4.

Group	Itself	Every proper subgroup cyclic?
K_4	abelian, not cyclic	yes
Q_8	non-abelian	yes
S_3	non-abelian	yes
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$	abelian, not cyclic	no (contains K_4)

4. Exam shortcuts and standard counter-examples

Exam shortcut. Subgroups of prime order p: each is generated by any of its $p-1$ non-identity elements, so their number is (elements of order p)/$(p-1)$. Example: S_4 has 8 three-cycles, hence 4 subgroups of order 3.

Exam shortcut. In $\mathbb{Z}_n$: order of $\langle\bar{k}\rangle$ is $n/\gcd(n,k)$, its index is $\gcd(n,k)$; the subgroup of order d is $\langle\overline{n/d}\rangle$; elements of order d: $\varphi(d)$ if $d \mid n$, else 0; solutions of $mx=0$: $\gcd(m,n)$. In $\mathbb{C}^*$ there are exactly $\varphi(n)$ elements of order n.

Exam shortcut. $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic iff $\gcd(m,n)=1$; $U(n)$ is cyclic iff $n = 1, 2, 4, p^k, 2p^k$ (p an odd prime).

Exam shortcut. Standard traps: $H \cup K$ is a subgroup only if one contains the other; $\langle(12)\rangle\langle(13)\rangle$ in S_3 has 4 elements, so HK need not be a subgroup; A_4 has no subgroup of order 6 although $6 \mid 12$; for infinite subsets closure is not enough ($\mathbb{N} \subset \mathbb{Z}$).

Exam shortcut. A group all of whose proper subgroups are cyclic need not be cyclic (K_4, Q_8, S_3). $(\mathbb{Q},+)$ is not cyclic, not even finitely generated, but every finitely generated subgroup of $\mathbb{Q}$ is cyclic. Every infinite group has infinitely many subgroups; every group of composite order has a proper non-trivial cyclic subgroup. A cyclic group with exactly one generator has at most 2 elements, since $\varphi(n)$ is even for $n \geq 3$.

5. Solved examples

Example 1. In $\mathbb{Z}_{36}$ find (a) the number of subgroups, (b) the number of generators, (c) the subgroup of order 9, (d) the number of elements of order 6, (e) the index of $\langle\overline{20}\rangle$.

Solution. (a) $36 = 2^2 3^2$, so $\tau(36) = 3 \cdot 3 = 9$. (b) $\varphi(36) = 36 \cdot \frac{1}{2} \cdot \frac{2}{3} = 12$. (c) $\langle\overline{36/9}\rangle = \langle\bar{4}\rangle = \{0, 4, 8, \ldots, 32\}$. (d) $\varphi(6) = 2$ (namely $\bar{6}, \overline{30}$). (e) $\gcd(36,20) = 4$, so $|\langle\overline{20}\rangle| = 9$ and the index is 4.

Example 2. Find the number of subgroups of $\mathbb{Z}_5 \times \mathbb{Z}_5$, and the number of its subgroups of order 5.

Solution. Every non-identity element has order 5, so there are 24 elements of order 5 and $24/4 = 6$ subgroups of order 5 ($= p+1$). Adding the trivial subgroup and the whole group: $6 + 2 = 8 = p+3$ subgroups.

Example 3. Find the number of subgroups, and of cyclic subgroups, of the dihedral group D_8 of order 16.

Solution. Rotation subgroups: one for each divisor of 8, $\tau(8) = 4$. Subgroups containing reflections: $\langle r^d, sr^i\rangle$ for $d \mid 8$, $0 \leq i < d$, which number $\sigma(8) = 1+2+4+8 = 15$. Total $4 + 15 = 19$. Cyclic ones: the 4 rotation subgroups plus the 8 subgroups $\{e, sr^i\}$, so $\tau(8) + 8 = 12$. (A brute-force check confirms 19 and 12.)

Example 4. Let $H, K \leq G$ with $|H| = 12$, $|K| = 18$, $|H \cap K| = 6$. Find $|HK|$. Can HK be a subgroup if $|G| = 54$?

Solution. $|HK| = \frac{12 \cdot 18}{6} = 36$. If $|G| = 54$, a subgroup of order 36 is impossible by Lagrange ($36 \nmid 54$), so HK is not a subgroup; in particular $HK \neq KH$ and neither H nor K is normal.

Example 5. Find the number of cyclic subgroups of $\mathbb{Z}_2 \times \mathbb{Z}_4$ and of D_4.

Solution. $\mathbb{Z}_2 \times \mathbb{Z}_4$ has 8 subgroups: $\{0\}$; three of order 2, generated by $(1,0), (0,2), (1,2)$; three of order 4: $\langle (0,1) \rangle$, $\langle (1,1) \rangle$ and the Klein group $\{(0,0), (1,0), (0,2), (1,2)\}$; and the whole group. The Klein subgroup and the whole group (no element of order 8) are not cyclic, so 6 are cyclic.
D_4: $\tau(4) + 4 = 3 + 4 = 7$ cyclic subgroups ($\{e\}$, $\langle r^2 \rangle$, $\langle r \rangle$ and four $\langle sr^i \rangle$), out of 10 subgroups.

Example 6. Which statements are true? (a) The centre of D_{12} (order 24) has order 2. (b) In a cyclic group of order 30, the number of x with $x^6 = e$ is 6. (c) $\mathbb{C}^*$ has exactly 4 elements of order 8. (d) $\{x \in \mathbb{R}^* : |x| \leq 1\}$ is a subgroup of $\mathbb{R}^*$.

Solution. (a) True: $n = 12$ is even, $Z(D_{12}) = \{e, r^6\}$. (b) True: $\gcd(6, 30) = 6$. (c) True: the primitive 8th roots of unity $e^{2\pi ik/8}$, $k = 1, 3, 5, 7$, number $\varphi(8) = 4$. (d) False: $\frac{1}{2}$ is in the set but its inverse 2 is not. Answer: (a), (b), (c).